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# SegEarth-R2: Towards Comprehensive Language-guided Segmentation for Remote Sensing Images

Zepeng Xin[1*], Kaiyu Li[1*], Luodi Chen[1], Wanchen Li[1], Yuchen Xiao[1], Hui Qiao[5], Weizhan Zhang[1], Deyu Meng[2,3,4], Xiangyong Cao[1,2†]
[1]Faculty of Electronic and Information Engineering, Xi'an Jiaotong University
[2]Ministry of Education Key Laboratory of Intelligent Networks and Network Security
[3]School of Mathematics and Statistics, Xi'an Jiaotong University
[4]Macao Institute of Systems Engineering, Macau University of Science and Technology
[5]China Telecom Shaanxi Branch

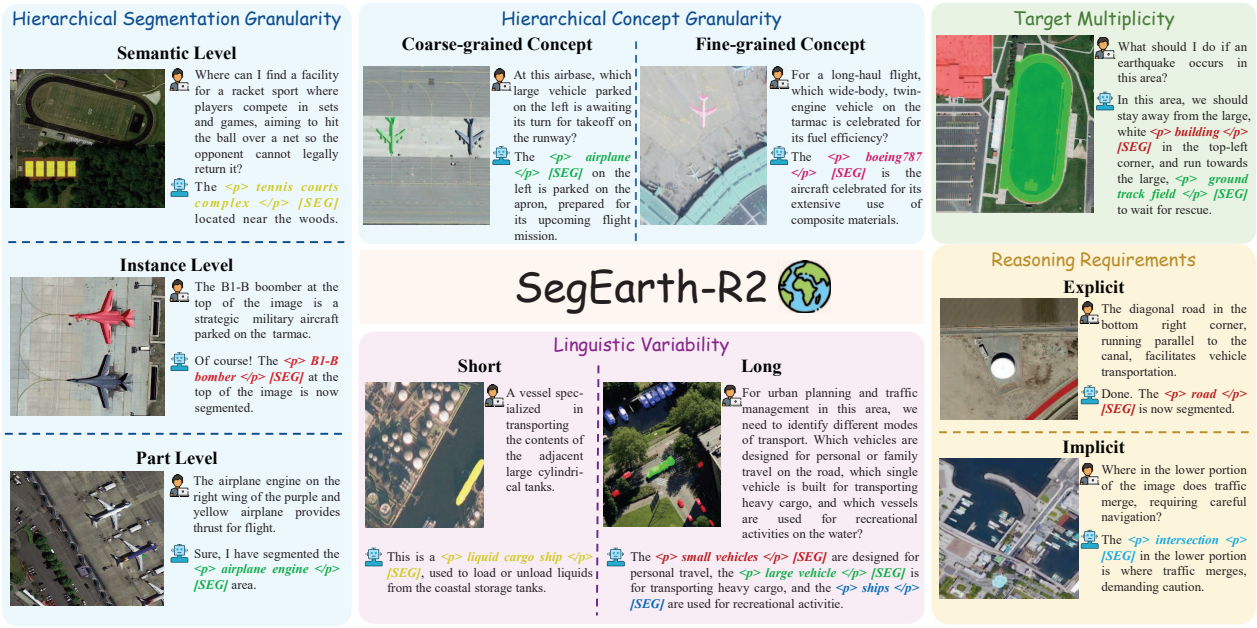



Figure 1. SegEarth-R2 can handle various language-guided segmentation in remote sensing across four key dimensions: (1) **Hierarchical Granularity**; (2) **Target Multiplicity**; (3) **Reasoning Requirements**; and (4) **Linguistic Variability**.

## Abstract

*Effectively grounding complex language to pixels in remote sensing (RS) images is a critical challenge for applications like disaster response and environmental monitoring. Current models can parse simple, single-target commands but fail when presented with complex geospatial scenarios, e.g., segmenting objects at various granularities, executing multi-target instructions, and interpreting implicit user intent. To drive progress against these failures, we present LaSeRS, the first large-scale dataset built for comprehensive training and evaluation across four critical dimensions of language-guided segmentation: hierarchical granularity, target multiplicity, reasoning requirements, and linguistic variability. By capturing these dimensions, LaSeRS moves beyond simple commands, providing a benchmark for complex geospatial reasoning. This addresses a critical gap: existing datasets oversimplify, leading to sensitivity-prone real-world models. We also propose SegEarth-R2, an MLLM architecture designed for comprehensive language-guided segmentation in RS, which directly confronts these challenges. The model's effectiveness stems from two key improvements: (1) a spatial attention supervision mechanism specifically handles the localization of small objects and their components, and (2) a flexible and efficient segmentation query mechanism that handles*

*Equal contribution
†Corresponding author (caoxiangyong@mail.xjtu.edu.cn)

*both single-target and multi-target scenarios. Experimental results demonstrate that our SegEarth-R2 achieves outstanding performance on LaSeRS and other benchmarks, establishing a powerful baseline for the next generation of geospatial segmentation. All data and code will be released at https://github.com/earth-insights/SegEarth-R2.*

## 1. Introduction

Imagine you are walking in a park when the ground begins to shake violently—an earthquake. Immediately, instinct kicks in and your mind swiftly translates the abstract goal of "finding a safe refuge" into a complex, spatially-aware plan (*e.g.*, the target multiplicity case depicted in Figure 1): simultaneously identifying the nearby, open ***ground track field*** (the safe destination) while also urgently moving away from the surrounding ***tall, swaying buildings*** (the hazards). This remarkable human ability to ground a high-level, compound instruction into a precise, multi-target, pixel-level action is vital for survival. Yet, it is precisely this ability that even advanced AI systems for Earth observation struggle to replicate. While these systems excel at executing literal commands [9, 23, 25, 27, 28, 36, 41, 50, 72, 79], they fall short when interpreting abstract natural language that requires understanding spatial relationships between multiple targets, especially in the context of precise language-guided segmentation in complex, real-world remote sensing (RS) environments.

This inability to interpret complex instructions directly hinders the deployment of advanced geoinformation systems. Indeed, language-guided segmentation holds immense promise across diverse and complex real-world RS applications, including environmental dynamics, urban planning, and, critically, disaster management [1, 3, 7, 26, 39, 45, 49, 58]. However, its development has been significantly inhibited by the lack of comprehensive training and evaluation resources. Specifically, existing datasets do not adequately capture the full range of required complexity, such as segmenting objects across hierarchical granularity and handling complex target multiplicity instructions.

To address these critical gaps, we introduce LaSeRS, a large-scale dataset of over 40k high-quality masks and 30k QA pairs, generated via a scalable semi-automated annotation pipeline. As illustrated in Figure 1, LaSeRS is constructed to serve as a comprehensive training and evaluation resource across the four dimensions of RS language-guided segmentation. Unlike prior works limited to single-target and explicit queries, LaSeRS systematically incorporates: **Hierarchical Granularity**, requiring both conceptual (coarse- to fine-grained) and segmentation (semantic to part-level) precision; **Target Multiplicity**, where a single instruction demands grounding of multiple objects; **Reasoning Requirements**, both explicit descriptions and challenging implicit instructions; and **Linguistic Variability**, spanning concise short queries and highly detailed long descriptions. Further, LaSeRS significantly pushes scale, offering more than 5× the category coverage of existing RS datasets [9, 28, 29, 36, 79], thereby significantly expanding the boundary of language-guided segmentation in RS.

The LaSeRS dataset poses some critical challenges to existing models [19, 28, 50, 62], motivating the introduction of our SegEarth-R2 model. We built our model to overcome two primary failures. First, existing methods struggle with fine-grained segmentation of small and part-level object in LaSeRS. SegEarth-R2 confronts this with a spatial attention supervision mechanism, enforcing an internal focus on precise details. Second, grounding multiple targets from a single instruction is inefficient or unsupported by previous query designs [28]. SegEarth-R2 address this with a flexible segmentation query mechanism capable of dynamically handling both single and multi-target requests.

In summary, our contributions are as follows:

- We introduce LaSeRS, the first benchmark for training and evaluation comprehensive language-guided segmentation in RS across four critical dimensions spanning hierarchical granularity, target multiplicity, reasoning requirements, and linguistic variability.
- We propose SegEarth-R2, a effective MLLM architecture for comprehensive language-guided segmentation in RS. It incorporates a spatial attention supervision for precise fine-grained localization and a flexible query mechanism for handling multi-target instructions.
- SegEarth-R2 sets a new state-of-the-art on LaSeRS and other public benchmarks. With only 3B parameters, it demonstrates superior performance against larger models, establishing a powerful yet highly efficient baseline for the comprehensive language-guided segmentation.

## 2. Related works

**Multimodal Large Language Models (MLLMs).** MLLMs build upon Large Language Models (LLMs) to acquire vision capabilities. In the general domain, pioneers like LLaVA [24, 32, 33] and InstructBLIP [6] aligned visual features with language representations using a vision-language connector, enhanced by instruction tuning. While these initial models focused on image-level understanding, subsequent works like GPT4RoI [75] and RegionGPT [10] advanced this by introducing regional understanding. This evolution is mirrored in the RS domain, where MLLMs have evolved from foundational conversational agents like RSGPT [14]. Subsequent RS-specific models have integrated more complex functionalities. For instance, GeoChat [18] supports region-specific inputs and visual grounding through oriented bounding box coordinates in its responses. Others extend capabilities to RS-video captioning, such as SkyEyeGPT [73], or integrate

Table 1. Comparison between LaSeRS and other related datasets. TI: Textual Instruction, TA: Textual Answer, BBox: Bounding Box, Seg: Segmentation Mask. LaSeRS outperforms other datasets with the largest number of classes and unique support for **Hierarchical Granularity**, **Target Multiplicity**, **Reasoning Requirements**, and **Linguistic Variability**.

| Dataset | Masks | Class Num | Annotation Type | Hierarchical Granularity | | Target Multiplicity | | Reasoning Requirements | | Linguistic Variability | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | Segmentation | Concept | Single | Multiple | Explicit | Implicit | Long | Short |
| EarthVQA [57] | 6,000 | 14 | TI, TA | ✗ | ✗ | ✓ | ✓ | ✓ | ✓ | ✗ | ✓ |
| RegSegRS [72] | 4,420 | 14 | TI, Seg | ✗ | ✗ | ✓ | ✗ | ✓ | ✗ | ✗ | ✓ |
| RRSIS-D [36] | 17,420 | 20 | TI, Seg, BBox | ✗ | ✗ | ✓ | ✗ | ✓ | ✗ | ✗ | ✓ |
| RISBench [9] | 52,472 | 26 | TI, Seg, BBox | ✗ | ✗ | ✓ | ✗ | ✓ | ✗ | ✗ | ✓ |
| EarthReason [28] | 5,434 | 28 | TI, TA, Seg | ✗ | ✗ | ✓ | ✗ | ✗ | ✓ | ✓ | ✗ |
| LaSeRS | 40,396 | 122 | TI, TA, Seg, BBox | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |

multi-sensor tasks, as exemplified by EarthGPT [76] and EarthDial [52]. However, despite these advances in interpretation [2, 15, 30, 42] and relation reasoning [43, 55, 73], a fundamental limitation persists: these models lack crucial pixel-level understanding and segmentation capabilities.

**Language-guided Segmentation.** Language-guided segmentation refers to locating target regions at the pixel level from natural language, including referring segmentation [8, 12, 34, 59, 65, 67, 81] and reasoning segmentation [16, 19, 48, 81]. In RS, RefSegRS [72] introduced the first referring segmentation dataset, while RRSIS-D [36] and some subsequent works [9, 41, 41, 68, 79, 79] expanded dataset scale and enriched prompts, though still limited to explicit category queries. EarthReason [28] further introduced geospatial pixel reasoning with the first reasoning segmentation dataset for Earth observation. Yet, while a valuable step, it primarily focuses on implicit reasoning for single targets, leaving the complex challenges of multi-granularity (*e.g.*, part-level and fine-grained concept) and multi-target grounding unaddressed. To bridge this gap, LaSeRS offers a more comprehensive benchmark that systematically incorporates all of these critical dimensions.

**MLLMs for language-guided segmentation.** Recent work has explored leveraging the reasoning capabilities of MLLMs for complex language-guided segmentation. In the general domain, a dominant approach employs MLLMs as a reasoning engine to provide prompt embeddings (*e.g.*, [SEG] tokens) for SAM [17, 47], thereby integrating reasoning into mask generation [11, 16, 19, 60, 64, 66, 71]. Other methods follow pipelines inspired by Mask2Former [4] ("propose-then-select") [56, 61, 62, 65, 77] or treat segmentation as text generation [20, 21]. However, these general-purpose methods often underperform on RS datasets, which are characterized by the prevalence of small objects and significant scale variations. This has spurred the development of specialized MLLMs for the RS field [37, 40, 41, 51, 69, 79], with methods like GeoPixel [50] adapting grounded conversation, SegEarth-R1 [28] and LISAt [44] handling implicit queries, Geo-R1 [78] and RemoteReasoner [70] exploring reinforcement learning. Despite these advances in RS-specific models, they still lack the ability to handle language-guided segmentation in complex scenarios, such as multi-target settings or multi-granularity (*e.g.*, part-level).

## 3. LaSeRS Dataset

### 3.1. Data Generation and Filtering

We develop a four-stage semi-automatic data generation pipeline to generate question-answer-mask triples, as shown in Figure 2. More details about the dataset construction can be found in Suppl. 7.

**Stage 1: Masks Generation.** Our mask annotations are sourced from two complementary workflows. First, we incorporate masks from the existing SAMRS [54] dataset. However, these annotations exhibit varying quality and are limited to ∼60 categories. Second, to expand coverage and introduce part-level details, we use SAM [17] to generate masks for the entire image using a grid of point prompts. Then, for common RS object categories like airplanes and ships, we cropped the object region and apply a denser grid of prompts within that region to produce more refined, part-level masks. This stage generates a large volume of coarse masks that require subsequent filtering.

**Stage 2: Masks Filtering.** To ensure annotation quality, we apply distinct filtering methods based on the mask source. For masks originating from SAMRS, we discard samples where the number of connected components in a mask does not match the number of provided bounding box annotations. Given that objects in remote sensing images often exhibit clear geometric properties (*e.g.*, the pronounced symmetry of airplanes and the rotational invariance typical of intersections), we then conduct geometric feature matching to filter out candidates that deviate significantly from a curated set of ideal, category-specific reference shapes (details in the Suppl. 7.3). In contrast, all masks generated by the second workflow are manually reviewed. Annotators assess each mask based on two criteria: semantic coherence, to ensure it represents a meaningful object, and high boundary quality, to ensure mask precision. This stage yields approximately 50K high-quality, semantically meaningful masks at varying segmentation granularities, from semantic- and instance-level to part-level.

**Stage 3: QAs Generation.** We used Gemini-2.5-pro [5]

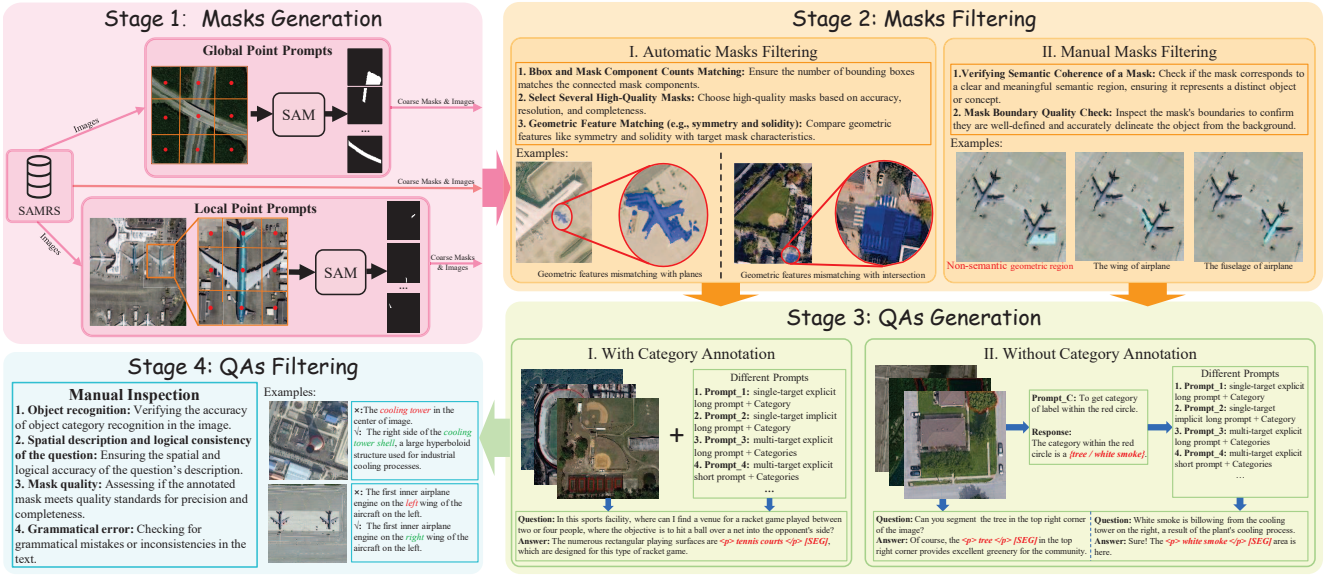
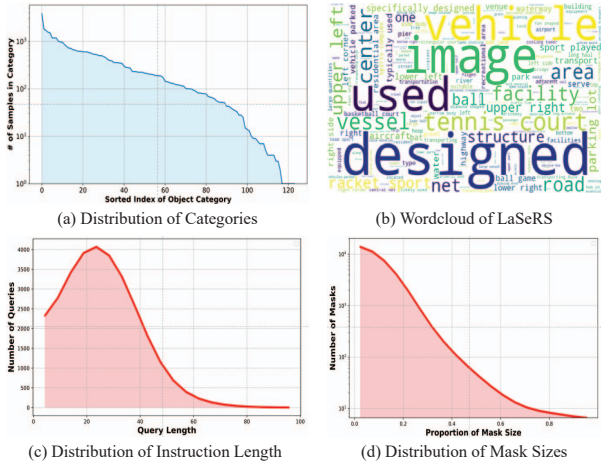



Figure 2. Overview of the construction of LaSeRS. Our scalable, semi-automatic annotation pipeline comprises four stages: two dedicated to data generation and two dedicated to filtering. Data quality is rigorously ensured through both automatic and manual verification.




(a) Distribution of Categories (b) Wordcloud of LaSeRS

(c) Distribution of Instruction Length (d) Distribution of Mask Sizes

Figure 3. Data Statistics of LaSeRS.

to create question-answer pairs, with the process differing based on the mask source. For masks with pre-existing category annotations from SAMRS, we leverage this information to guide the generation of diverse textual annotations. Conversely, for unlabeled masks from our second workflow, we first employ a prompting strategy to assign a category to each mask. This assigned category subsequently guides the generation of varied textual annotations. In both cases, we utilize multiple prompt templates to create question-answer pairs of varying styles, including explicit vs. implicit and detailed vs. concise. This stage ultimately yields approximately 40K question-answer pairs. However, due to potential hallucinations from the Gemini-2.5-pro, these pairs require further filtering.

**Stage 4: QAs Filtering.** Finally, all generated question-answer pairs are manually reviewed. This review verifies the accuracy of object recognition, the logical consistency of spatial descriptions, the quality of the associated masks, and checks for grammatical errors to ensure the reliability of the final dataset. This final filtering stage yields 30,830 high-quality question-answer-masks triples. Additionally, to accommodate the demand for coarse localization, we convert the masks into bounding box annotations via mask-to-bbox [31].

### 3.2. Data Statistics

As detailed in Table 1, LaSeRS is a large-scale dataset comprising 40,396 high-quality masks. The dataset's annotations span four types: textual instructions, textual answers, segmentation masks, and bounding boxes for coarse localization. The test set consists of 1,900 held-out masks and is meticulously partitioned to evaluate performance across four critical dimensions: **Hierarchical Segmentation Granularity** (semantic, instance, and part-level), **Target Multiplicity** (single, multiple), **Reasoning Requirements** (explicit, implicit), and **Linguistic Variability** (short, long). Figure 3 summarizes the statistical properties of LaSeRS. Panel (a) shows the distribution across 122 object categories, highlighting our dataset's broad coverage. This ensures the QA pairs are not skewed towards specific categories and exhibit high diversity. Further details on the categories are provided in the Suppl. 7.6. Panel (b) presents a word cloud of the instruction vocabulary. Panel (c) illustrates the instruction length distribution, ranging from just a few words to over 80, ensuring linguistic variability. Fi-

nally, panel (d) highlights the distribution of mask sizes, with a significant scale range from small, fine-grained parts to large regions exceeding 80% of the image area.

## 4. SegEarth-R2

To address the challenges of language-guided segmentation in RS, we introduce SegEarth-R2, a framework designed for precise, language-driven segmentation in RS images. As depicted in Figure 4, the model is composed of two primary components. The first is an MLLM that processes the input image and textual instruction to perform reasoning and generate a textual response. The second is a dedicated segmentation head that interprets the MLLM's internal semantic representation, specifically the [SEG] token queries, to produce pixel-level masks. The performance of SegEarth-R2 relies on two key mechanisms corresponding to the above components: a spatial attention supervision mechanism applied to the model's internal attention and a flexible and efficient segmentation query mechanism for mask generation, which are detailed in the following.

### 4.1. Spatial Attention Supervision

A defining characteristic of RS images is the extreme variation in object scales, where targets can range from vast geographical features to entities spanning only a few pixels (*e.g.*, the small engine of an airplane). This presents a significant challenge for standard training paradigms. Supervision that relies solely on the final segmentation mask proves insufficient for instilling precise spatial awareness, as the learning signal becomes severely diluted when backpropagated to the network's shallower layers. This dilution effect often leads to poor localization of small or complexly shaped objects, as visualized in Figure 4(d).

To counteract this, we introduce a spatial attention supervision mechanism, which provides direct supervisory signals to the model's internal vision-language representations. Rather than inferring focus from the final output, our approach constitutes a direct intervention within the model's reasoning pathway, compelling it to distinguish between foreground and background regions at the attention level. This process is illustrated in Figure 4(c).

This method operates on the attention maps originating from the [SEG] token to the image patch tokens within the MLLM's transformer layers. Let $A_{(m,n)} \in \mathbb{R}^{d\times d}$ denote the attention map for the $n$-th head in the $m$-th transformer block, where the model has a total of $M$ blocks and $N$ heads per block, $d \times d$ represents the spatial dimensions of the image feature grid to which attention is applied. We then aggregate these individual maps into a unified attention grid by averaging them across all layers and heads, *i.e.*, $A_S = \frac{1}{MN}\sum_{m=1}^{M}\sum_{n=1}^{N} A_{(m,n)}$. Using a downsampled version of the ground truth mask, $\hat{G} \in \{0,1\}^{d\times d}$, we calculate the average attention score, $a$, across all background regions:

$$a = \frac{\sum_{i,j} A_S(i,j)\cdot(1-\hat{G}(i,j))}{\sum_{i,j}(1-\hat{G}(i,j))}$$

where $A_S(i,j)$ and $\hat{G}(i,j)$ represent the $i$-th row and $j$-th column of the corresponding tensors.

The spatial attention supervision loss, $\mathcal{L}_S$, is then formulated to enforce a maximal separation between the attention scores on the foreground regions and the calculated background average $a$:

$$\mathcal{L}_S = -\log\left(\frac{\sum_{i,j}(A_S(i,j)-a)^2\cdot\hat{G}(i,j)}{\sum_{i,j}\hat{G}(i,j)}\right)$$

By maximizing the dissimilarity, this loss directly encourages the model to sharpen its focus specifically on the target area. This provides a clear and localized learning signal to the intermediate layers, substantially improving the model's overall ability to ground and segment challenging objects with remarkable high fidelity.

### 4.2. Segmentation Query Mechanism

Previous approaches, such as InstructSeg [62], rely on a cumbersome "propose-then-select" strategy. They first generate a large pool of candidate masks (*e.g.*, 100) and are subsequently plagued by a redundant and time-consuming matching process to filter for the desired output; this design is not only computationally inefficient but also inflexible. On the other hand, while SegEarth-R1's [28] "instruction-as-query" method offers simplicity, it is inherently limited by its assumption of a single target per instruction. This naive one-to-one mapping makes it fundamentally unsuitable for complex, multi-target scenarios.

As illustrated in Figure 4, to handle multi-target scenarios, we adopt the approach from LISA [19] by introducing a special [SEG] token. The model learns to dynamically output multiple [SEG] tokens based on the instruction's context (*e.g.* "*In this area, we should stay away from the large, white building [SEG] in the top-left corner, and run towards the large, ground track field [SEG] to wait for rescue.*"), which then function as individual segmentation queries. Our architecture is composed of a Hierarchical Visual Encoder, *i.e.*, the Swin-B [38] which extracts multi-level image features. These features are then processed by a Pixel Decoder and a Transformer Decoder; we note that both of these modules adopt the identical architecture from Mask2Former [4]. The Pixel Decoder fuses the hierarchical features from the Swin-B, and the Transformer Decoder utilizes the [SEG] token queries to interact with these fused features, generating the final masks. This flexible and efficient segmentation query mechanism not only evades the redundant matching of "propose-then-select" methods but also capably handles the complex multi-target scenarios where "instruction-as-query" methods fail.

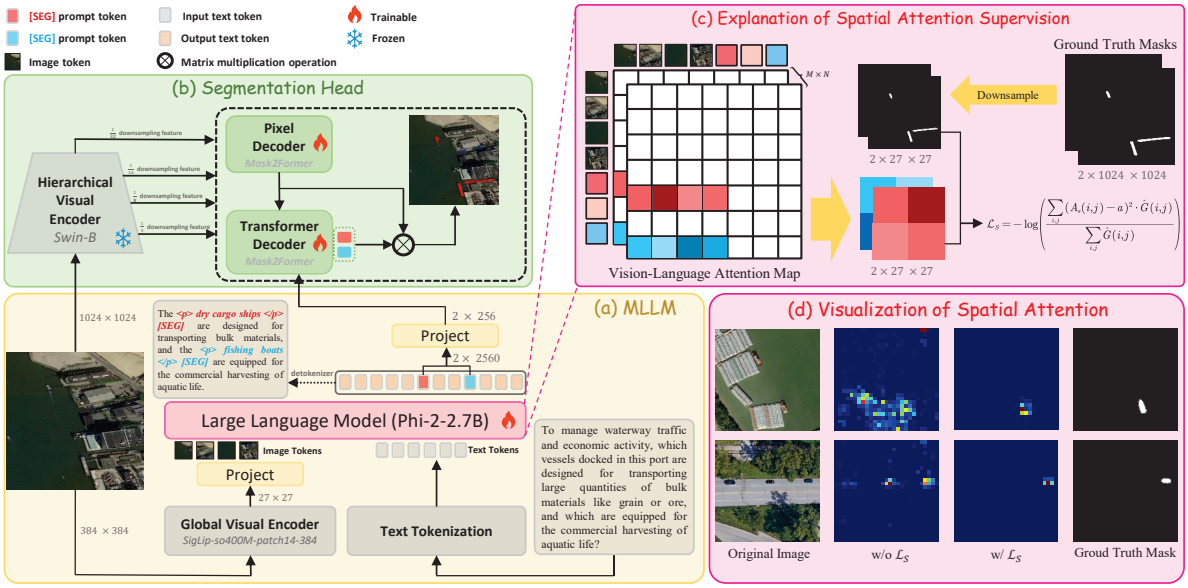



Figure 4. The overall architecture of SegEarth-R2. (a) is an MLLM for interpreting the image and instruction, (b) is the segmentation head, which includes a hierarchical visual encoder for generating multi-scale image features, and two decoders for producing masks (mask decoders). (c) illustrates the formation process of the spatial attention supervision. (d) presents a visualization of the attention map from the 6th head of the 16th layer in the MLLM, with additional visual results in the Suppl. 9.1

Table 2. Performance comparison on LaSeRS. Baselines including LISA, PixelLM, $M^2$A, and GeoPixel are fine-tuned directly on LaSeRS. GLaMM-ft is pre-trained on large-scale natural image segmentation datasets before fine-tuning. Metrics are gIoU/cIoU. **Bold** and underlined indicate the best and second-best scores, respectively.

| Model | Hierarchical Segmentation Granularity | | | Target Multiplicity | | Reasoning Requirements | | Linguistic Variability | | Avg. |
|---|---|---|---|---|---|---|---|---|---|---|
| | Semantic | Instance | Part | Single | Multiple | Explicit | Implicit | Short | Long | |
| LISA-7B [19] | 26.4/23.2 | 20.5/25.0 | 16.1/11.6 | 37.3/32.2 | 18.2/22.4 | 27.1/24.3 | 21.5/25.6 | 34.1/27.8 | 38.4/33.9 | 26.6/25.1 |
| LISA-13B [19] | 27.0/24.5 | 22.3/25.6 | 17.7/13.1 | 38.4/34.2 | 19.9/23.5 | 27.1/25.5 | 22.6/25.8 | 35.2/28.0 | 38.4/34.3 | 27.6/26.1 |
| PixelLM-7B [48] | 32.0/32.8 | 26.6/30.0 | 13.2/16.5 | 44.3/40.4 | 20.2/23.5 | 25.0/23.1 | 23.9/21.9 | 41.6/38.9 | 37.1/34.5 | 29.3/29.1 |
| PixelLM-13B [48] | 31.6/34.0 | 27.5/30.2 | 15.8/17.6 | 42.2/40.5 | 20.9/22.4 | 26.3/24.4 | 25.9/22.1 | 42.0/39.1 | 37.1/34.5 | 29.9/29.4 |
| GLaMM-ft-7B [46] | 44.8/47.9 | 41.2/48.3 | 32.6/42.7 | 47.3/50.3 | 32.2/41.0 | 59.1/60.3 | 42.6/44.8 | 50.4/54.8 | 42.6/44.8 | 43.6/48.3 |
| $M^2$A-7B [16] | 30.1/33.0 | 23.0/24.8 | 18.6/17.2 | 45.4/37.6 | 20.9/24.8 | 35.8/30.4 | 23.3/26.7 | 35.8/32.8 | 41.5/36.7 | 30.5/29.3 |
| GeoPixel-8B [50] | 51.4/57.2 | 44.1/49.3 | 43.9/52.4 | 55.0/45.8 | **49.2**/49.7 | 66.5/61.3 | 41.1/58.3 | 51.1/59.3 | **51.4**/63.2 | 50.4/55.2 |
| SegEarth-R2-3B | **60.2/71.8** | **65.4/70.3** | **64.8/68.3** | **55.1/69.2** | 38.3/**56.2** | **78.4/80.4** | **42.8/59.7** | **60.2/69.9** | 50.1/**65.7** | **57.2/67.9** |

# 5. Experiment

## 5.1. Implementation Details

**Training objectives.** We train SegEarth-R2 with a unified loss function $\mathcal{L}$, which is the weighted sum of four distinct components:

$$\mathcal{L} = \mathcal{L}_t + \mathcal{L}_b + \mathcal{L}_d + \lambda_S \mathcal{L}_S$$

Here, $\mathcal{L}_t$ is the standard autoregressive cross-entropy loss for text generation. For mask supervision, we combine the per-pixel binary cross-entropy loss $\mathcal{L}_b$ and DICE loss $\mathcal{L}_d$, while $\mathcal{L}_S$ denotes the spatial attention supervision, which provides direct supervision on the model's internal attention and is controlled by the coefficient $\lambda_S$.

**Experiment Settings.** We use the pre-trained lightweight MLLM Mipha-3B [80] as our base model. Following SegEarth-R1 [28], we use Swin-B [38] as the hierarchical visual encoder and the mask decoder from Mask2Former [4]. Furthermore, to conserve GPU memory, we keep all Visual Encoders frozen and employ LoRA with a rank of 8 to fine-tune the LLM, while the smaller Pixel Decoder and Transformer Decoder components are fully fine-tuned. More training details can be found in Suppl. 8.1.

**Metrics.** We use gIoU and cIoU as evaluation metrics, following prior studies [19, 63].

## 5.2. Main Results

As shown in Table 2, we conduct a comprehensive and fair evaluation on the LaSeRS benchmark, analyzing performance across four key dimensions.

Table 3. Performence comparison on three RS referring segmentation benchmarks. To ensure a fair comparison, our results are obtained by training separately on each of the three training set. The metric is gIoU.

| Method | Pub | RRSIS-D | | RefSegRS | | RISBench | |
|---|---|---|---|---|---|---|---|
| | | val | test | val | test | val | test |
| *Segmentation Specialists* | | | | | | | |
| CMPC+ [35] | TPAMI'21 | 51.4 | 50.2 | 47.1 | 43.7 | 45.8 | 46.7 |
| RIS-DMMI [13] | CVPR'23 | 60.7 | 60.1 | 65.7 | 52.2 | 62.6 | 63.9 |
| LAVT [67] | CVPR'22 | 61.5 | 61.0 | 61.5 | 47.4 | 60.5 | 61.9 |
| RMSIN [36] | CVPR'24 | 65.1 | 64.2 | 73.8 | 62.6 | 61.8 | 63.1 |
| *MLLM based segmentation* | | | | | | | |
| LISA [19] | CVPR'24 | 27.8 | 26.8 | - | - | - | - |
| PixelLM [48] | CVPR'24 | 33.9 | 31.7 | - | - | - | - |
| GeoGround [79] | arXiv'24 | 61.1 | 60.5 | - | - | - | - |
| GeoPixel [50] | ICML'25 | 68.0 | 67.3 | - | - | - | - |
| SegEarth-R1 [28] | arXiv'25 | 67.6 | 66.4 | 82.2 | 72.5 | - | - |
| Text4Seg++ [22] | arXiv'25 | 64.1 | 62.8 | - | - | - | - |
| SegEarth-R2 | | **68.8** | **67.9** | **84.4** | **74.8** | **69.8** | **70.5** |

Table 4. Performence comparison on RS reasoning segmentation benchmark: EarthReason [28]. Our model is trained solely on the EarthReason training set.

| Method | LLM | Val | | Test | | Avg. |
|---|---|---|---|---|---|---|
| | | gIoU | cIoU | gIoU | cIoU | |
| LISA [19] | Vicuna-7B | 61.0 | 57.4 | 60.9 | 59.1 | 59.6 |
| PixelLM [48] | Vicuna-7B | 57.9 | 57.8 | 60.0 | 59.2 | 58.7 |
| SegEarth-R1 [28] | Phi-1.5-1.3B | 68.6 | 64.1 | 70.8 | 68.3 | 68.0 |
| RemoteReasoner [70] | Qwen2.5-7B | 69.0 | 67.8 | 71.0 | 69.1 | 69.2 |
| Text4Seg++ [22] | Qwen2-7B | 71.9 | **69.8** | 73.0 | 65.6 | 70.1 |
| SegEarth-R2-3B | Phi-2-2.7B | **72.3** | 68.1 | **73.5** | **69.5** | **70.9** |

**Hierarchical Segmentation Granularity.** We observe a consistent performance degradation across nearly all models as the segmentation granularity refines, *i.e.*, from the semantic-level, to the instance-level, and finally to the part-level, which often spans only a few pixels. This trend is expected, as finer granularities necessitate more precise model localization. Benefiting from our proposed spatial attention supervision mechanism, our model achieves the best performance across all granularities. This superiority is particularly pronounced at the challenging part-level, where our model surpasses the second-best method by a significant 20-point margin.

**Target Multiplicity.** For complex multi-target scenarios, we observe a significant performance degradation across most models. GeoPixel is a notable exception, which shows much less degradation, a result we attribute to the inherent strength of its 8B MLLM foundation [74]. We hypothesize that our model's 3B size limits its performance in these complex multi-target scenarios. Even so, our model still secures the best performance on single-target tasks and the second-best on multi-target tasks, significantly outperforming other 7B and even 13B models.

**Reasoning Requirements.** All models exhibit significantly better performance on explicit instructions compared to implicit ones. This is an expected outcome, as implicit instructions, unlike their explicit counterparts, require the model to infer the target segmentation region by reasoning over the instruction's context and its internal geographic knowledge. Notably, our model achieves excellent performance on both explicit and implicit instructions.

**Linguistic Variability.** Regarding linguistic variability, we observe an interesting phenomenon: models respond differently to instruction length. Specifically, LISA, $M^2$A, and GeoPixel excel at processing long, detailed instructions. In contrast, PixelLM, GLaMM, and SegEarth-R2 show more prominent performance on short, concise instructions. We hypothesize this divergence stems from fundamental differences in their model architectures. Notably, SegEarth-R2 demonstrates strong robustness, achieving superior performance on both instruction types.

**Overall.** With only 3B parameters, SegEarth-R2 significantly outperforms all other models on average, a result that showcases the superiority of our architecture. More qualitative results can be found in the Suppl. 10.1.

### 5.3. Results on Existing Benchmarks

To ensure SegEarth-R2's effectiveness is not limited to our proposed benchmark, we assess its generalization capabilities on established public datasets. For fair comparison, the model was trained separately on each benchmark's official training split.

**RS Referring Segmentation.** We evaluate on three referring segmentation benchmarks: RRSIS-D [36], RefSegRS [72], and RISBench [9]. As shown in Table 3, SegEarth-R2 demonstrates strong cross-dataset generalization. On the RRSIS-D dataset, it outperforms the powerful 8B model, GeoPixel. On RefSegRS, SegEarth-R2 surpasses the previous state-of-the-art method by over 2 points. Furthermore, our model achieves the best performance on the larger-scale RISBench dataset. Qualitative results are provided in the Suppl. 10.2.

**RS Reasoning Segmentation.** On the more challenging EarthReason [28] benchmark for implicit reasoning segmentation in RS, as shown in Table 4, our model achieves an average score of 70.9, outperforming the previous state-of-the-art, Text4Seg++ (70.1), which represents masks using text. It also surpasses RemoteReasoner, a method exploring reinforcement learning for RS reasoning segmentation, demonstrating the superiority of our architecture in processing complex implicit instructions. Qualitative results are provided in Suppl. 10.3.

### 5.4. Ablation Study

We conduct thorough ablation studies on our proposed spatial attention supervision ($\mathcal{L}_S$) and the adopted segmentation query mechanism. Furthermore, we investigate different combinations of the Segmentation Head (shown in Fig-

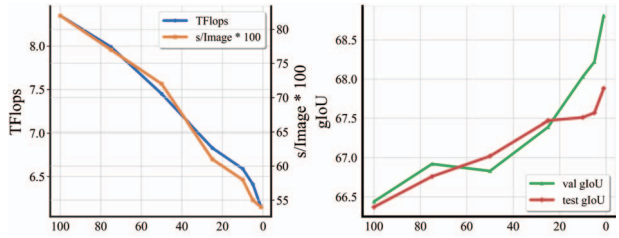



Figure 5. The left figure shows that as the number of segmentation queries decreases, both computational cost (TFLOPs) and inference time are correspondingly reduced. The right figure illustrates that this reduction in queries leads to a gradual increase in gIoU scores on the RRSIS-D validation and test sets.

Table 5. Ablation experiment on $\lambda_S$, which controls the strength of the spatial attention supervision in the MLLM inner layers. The metric is gIoU.

| $\lambda_S$ | $\mathcal{L}_S$ | RRSIS-D | | RefSegRS | | EarthReason | |
|---|---|---|---|---|---|---|---|
| | | val | test | val | test | val | test |
| 0 | × | 68.5 | 66.6 | 81.1 | 70.4 | 72.2 | 72.9 |
| 0.1 | ✓ | 68.1 | 67.3 | 82.2 | 71.1 | 70.1 | 71.8 |
| 0.05 | ✓ | 68.6 | 67.7 | 82.2 | 71.7 | 72.1 | 73.0 |
| 0.01 | ✓ | **68.8** | **67.9** | **82.4** | **71.8** | **72.3** | **73.5** |

Table 6. The comparison between different combination of segmentation head. ViT-H: Vision Transformer Huge, Swin-B: Swin Transformer Base, SAM: Segment Anything Model, M2F: Mask2Former. The metric is gIoU.

| Mask Decoder | Visual Encoder | EarthReason | | RefSegRS | |
|---|---|---|---|---|---|
| | | val | test | val | test |
| SAM [17] | ViT-H [53] | 62.7 | 65.4 | 73.9 | 60.3 |
| SAM 2 [47] | ViT-H [53] | 64.3 | 64.9 | 74.2 | 64.2 |
| M2F [4] | Swin-B [38] | **72.3** | **73.5** | **82.4** | **71.8** |

ure 4) to demonstrate the superiority of our architecture.

**Spatial Attention Supervision Effect.** The coefficient $\lambda_S$ controls the weight of spatial attention supervision $\mathcal{L}_S$. We perform an ablation study on $\lambda_S$, and the results are shown in Table 5. As shown, setting $\lambda_S = 0$, which removes this supervision, results in the worst performance, highlighting the necessity of spatial attention guidance. However, we observe that excessively large values of $\lambda_S$ (*e.g.*, 0.1 or 0.05) degrade performance. We hypothesize this occurs because overly constraining the MLLM's internal attention impairs its reasoning abilities. We find that $\lambda_S = 0.01$ strikes the optimal balance and adopt it as our final setting. Notably, this value represents the best result in our discrete ablation set; further fine-grained searches could potentially yield a marginally better optimum. The improvement from this supervision is small on the EarthReason benchmark, which we attribute to its limited scale variation and the predominance of large objects.

**Segmentation Query Mechanism Effect.** Previous works [56, 61, 62, 77] use 100 segmentation queries to generate candidate masks and select the best one via a matching algorithm. Our experiments show that this approach is both redundant and inefficient. Specifically, we evaluate the number of segmentation queries $k$, using values $k \in \{100, 75, 50, 25, 10, 3, 1\}$. When $k > 1$, the single [SEG] token is replaced by a sequence of $k$ independent query tokens, [SEG1] [SEG2] ... [SEG$k$]. The model correspondingly generates $k$ proposal masks, and a matching algorithm [4] is then employed to select the optimal mask from the $k$ candidates. When $k = 1$, the model generates a single mask that is directly used as the final output, bypassing the matching process. As shown in Figure 5, reducing the number of segmentation queries from 100 to just 1 results in a 34.1% reduction in inference time (from 0.82 to 0.54 s/image) and a 27.4% decrease in computational cost (from 8.4 to 6.1 TFLOPs), while also improving the gIoU score on the RRSIS-D [36] validation set by +2.4 points (66.4 → 68.8) and on the test set by +1.6 points (66.3 → 67.9). Visual analysis is provided in the Suppl. 9.2.

**Combination of Segmentation Head Analysis.** We substitute the Pixel Decoder and Transformer Decoder module (Figure 4) with SAM or SAM2 decoder. Concurrently, the Hierarchical Visual Encoder (Swin-B) is also replaced with the corresponding ViT-H backbone. Table 6 shows the results on EarthReason [28] and RefSegRS [72]. The combination of a multi-scale Swin Transformer and a Mask2Former decoder significantly outperforms architectures based on ViT-H and SAM or SAM2 decoder. This confirms that a hierarchical vision backbone, which is adept at capturing features across a wide range of scales, is better suited for the unique challenges of RS images compared to monolithic vision transformers. Our architectural choices are thus empirically validated as optimal for RS domain.

## 6. Conclusion

In this work, we explore comprehensive language-guided segmentation in RS. To advance this task, we present LaSeRS, the first large-scale benchmark designed to systematically investigate four key dimensions of language-guided segmentation: hierarchical granularity, target multiplicity, complex reasoning, and linguistic variability. This benchmark provides the community with a valuable resource to drive progress beyond simple, single-target instructions. To address the challenges of this benchmark, we propose SegEarth-R2, an effective and efficient MLLM architecture. Our evaluations show that SegEarth-R2 not only excels on the challenges posed by LaSeRS but also establishes a new SOTA across a range of established referring and reasoning segmentation benchmarks. Ultimately, we hope this work catalyzes a paradigm shift in the RS community, moving beyond parsing simple, single-target instructions toward developing comprehensive, unified language-guided segmentation.

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## References

[1] Xiao An, Jiaxing Sun, Zihan Gui, and Wei He. Choice: Benchmarking the remote sensing capabilities of large vision-language models. *arXiv preprint arXiv:2411.18145*, 2024. 2

[2] Yakoub Bazi, Laila Bashmal, Mohamad Mahmoud Al Rahhal, Riccardo Ricci, and Farid Melgani. Rs-llava: A large vision-language model for joint captioning and question answering in remote sensing imagery. *Remote Sensing*, 16(9): 1477, 2024. 3

[3] Zhenyuan Chen, Chenxi Wang, Ningyu Zhang, and Feng Zhang. Rscc: A large-scale remote sensing change caption dataset for disaster events. *arXiv preprint arXiv:2509.01907*, 2025. 2

[4] Bowen Cheng, Ishan Misra, Alexander G Schwing, Alexander Kirillov, and Rohit Girdhar. Masked-attention mask transformer for universal image segmentation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 1290–1299, 2022. 3, 5, 6, 8

[5] Gheorghe Comanici, Eric Bieber, Mike Schaekermann, Ice Pasupat, Noveen Sachdeva, Inderjit Dhillon, Marcel Blistein, Ori Ram, Dan Zhang, Evan Rosen, et al. Gemini 2.5: Pushing the frontier with advanced reasoning, multimodality, long context, and next generation agentic capabilities. *arXiv preprint arXiv:2507.06261*, 2025. 3

[6] Wenliang Dai, Junnan Li, Dongxu Li, Anthony Tiong, Junqi Zhao, Weisheng Wang, Boyang Li, Pascale N Fung, and Steven Hoi. Instructblip: Towards general-purpose vision-language models with instruction tuning. *Advances in neural information processing systems*, 36:49250–49267, 2023. 2

[7] Muhammad Sohail Danish, Muhammad Akhtar Munir, Syed Roshaan Ali Shah, Kartik Kuckreja, Fahad Shahbaz Khan, Paolo Fraccaro, Alexandre Lacoste, and Salman Khan. Geobench-vlm: Benchmarking vision-language models for geospatial tasks. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2025. 2

[8] Henghui Ding, Chang Liu, Suchen Wang, and Xudong Jiang. Vision-language transformer and query generation for referring segmentation. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 16321–16330, 2021. 3

[9] Zhe Dong, Yuzhe Sun, Tianzhu Liu, Wangmeng Zuo, and Yanfeng Gu. Cross-modal bidirectional interaction model for referring remote sensing image segmentation. *arXiv preprint arXiv:2410.08613*, 2024. 2, 3, 7

[10] Qiushan Guo, Shalini De Mello, Hongxu Yin, Wonmin Byeon, Ka Chun Cheung, Yizhou Yu, Ping Luo, and Sifei Liu. Regiongpt: Towards region understanding vision language model. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 13796–13806, 2024. 2

[11] Yue Han, Jiangning Zhang, Junwei Zhu, Runze Hou, Xiaozhong Ji, Chuming Lin, Xiaobin Hu, Zhucun Xue, and Yong Liu. Groundingface: Fine-grained face understanding via pixel grounding multimodal large language model. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pages 3942–3951, 2025. 3

[12] Rui Hu, Lianghui Zhu, Yuxuan Zhang, Tianheng Cheng, Lei Liu, Heng Liu, Longjin Ran, Xiaoxin Chen, Wenyu Liu, and Xinggang Wang. Groundingsuite: Measuring complex multi-granular pixel grounding. *arXiv preprint arXiv:2503.10596*, 2025. 3

[13] Yutao Hu, Qixiong Wang, Wenqi Shao, Enze Xie, Zhenguo Li, Jungong Han, and Ping Luo. Beyond one-to-one: Rethinking the referring image segmentation. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 4067–4077, 2023. 7

[14] Yuan Hu, Jianlong Yuan, Congcong Wen, Xiaonan Lu, Yu Liu, and Xiang Li. Rsgpt: A remote sensing vision language model and benchmark. *ISPRS Journal of Photogrammetry and Remote Sensing*, 224:272–286, 2025. 2

[15] Jeremy Andrew Irvin, Emily Ruoyu Liu, Joyce Chuyi Chen, Ines Dormoy, Jinyoung Kim, Samar Khanna, Zhuo Zheng, and Stefano Ermon. Teochat: A large vision-language assistant for temporal earth observation data. In *International Conference on Learning Representations*, 2025. 3

[16] Donggon Jang, Yucheol Cho, Suin Lee, Taehyeon Kim, and Daeshik Kim. MMR: A large-scale benchmark dataset for multi-target and multi-granularity reasoning segmentation. In *The Thirteenth International Conference on Learning Representations*, 2025. 3, 6

[17] Alexander Kirillov, Eric Mintun, Nikhila Ravi, Hanzi Mao, Chloe Rolland, Laura Gustafson, Tete Xiao, Spencer Whitehead, Alexander C Berg, Wan-Yen Lo, et al. Segment anything. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 4015–4026, 2023. 3, 8

[18] Kartik Kuckreja, Muhammad Sohail Danish, Muzammal Naseer, Abhijit Das, Salman Khan, and Fahad Shahbaz Khan. Geochat: Grounded large vision-language model for remote sensing. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 27831–27840, 2024. 2

[19] Xin Lai, Zhuotao Tian, Yukang Chen, Yanwei Li, Yuhui Yuan, Shu Liu, and Jiaya Jia. Lisa: Reasoning segmentation via large language model. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 9579–9589, 2024. 2, 3, 5, 6, 7

[20] Mengcheng Lan, Chaofeng Chen, Yue Zhou, Jiaxing Xu, Yiping Ke, Xinjiang Wang, Litong Feng, and Wayne Zhang. Text4seg: Reimagining image segmentation as text generation. *arXiv preprint arXiv:2410.09855*, 2024. 3

[21] Mengcheng Lan, Chaofeng Chen, Jiaxing Xu, Zongrui Li, Yiping Ke, Xudong Jiang, Yingchen Yu, Yunqing Zhao, and Song Bai. Text4seg++: Advancing image segmentation via generative language modeling. *arXiv preprint arXiv:2509.06321*, 2025. 3

[22] Mengcheng Lan, Chaofeng Chen, Jiaxing Xu, Zongrui Li, Yiping Ke, Xudong Jiang, Yingchen Yu, Yunqing Zhao, and Song Bai. Text4seg++: Advancing image segmentation via generative language modeling, 2025. 7

[23] Sen Lei, Xinyu Xiao, Tianlin Zhang, Heng-Chao Li, Zhenwei Shi, and Qing Zhu. Exploring fine-grained image-text alignment for referring remote sensing image segmentation. *IEEE Transactions on Geoscience and Remote Sensing*, 2024. 2

[24] Bo Li, Yuanhan Zhang, Dong Guo, Renrui Zhang, Feng Li, Hao Zhang, Kaichen Zhang, Peiyuan Zhang, Yanwei Li, Ziwei Liu, et al. Llava-onevision: Easy visual task transfer. *arXiv preprint arXiv:2408.03326*, 2024. 2

[25] Kaiyu Li, Xiangyong Cao, Yupeng Deng, Chao Pang, Zepeng Xin, Deyu Meng, and Zhi Wang. Dynamicearth: How far are we from open-vocabulary change detection? *arXiv preprint arXiv:2501.12931*, 2025. 2

[26] Kaiyu Li, Xiangyong Cao, Ruixun Liu, Shihong Wang, Zixuan Jiang, Zhi Wang, and Deyu Meng. Annotation-free open-vocabulary segmentation for remote-sensing images. *arXiv preprint arXiv:2508.18067*, 2025. 2

[27] Kaiyu Li, Ruixun Liu, Xiangyong Cao, Xueru Bai, Feng Zhou, Deyu Meng, and Zhi Wang. Segearth-ov: Towards training-free open-vocabulary segmentation for remote sensing images. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pages 10545–10556, 2025. 2

[28] Kaiyu Li, Zepeng Xin, Li Pang, Chao Pang, Yupeng Deng, Jing Yao, Guisong Xia, Deyu Meng, Zhi Wang, and Xiangyong Cao. Segearth-r1: Geospatial pixel reasoning via large language model. *arXiv preprint arXiv:2504.09644*, 2025. 2, 3, 5, 6, 7, 8

[29] Xiang Li, Jian Ding, and Mohamed Elhoseiny. Vrsbench: A versatile vision-language benchmark dataset for remote sensing image understanding. In *The Thirty-eight Conference on Neural Information Processing Systems Datasets and Benchmarks Track*. 2

[30] Zhenshi Li, Dilxat Muhtar, Feng Gu, Yanglangxing He, Xueliang Zhang, Pengfeng Xiao, Guangjun He, and Xiaoxiang Zhu. Lhrs-bot-nova: Improved multimodal large language model for remote sensing vision-language interpretation. *ISPRS Journal of Photogrammetry and Remote Sensing*, 227:539–550, 2025. 3

[31] Fan Liu, Delong Chen, Zhangqingyun Guan, Xiaocong Zhou, Jiale Zhu, Qiaolin Ye, Liyong Fu, and Jun Zhou. Remoteclip: A vision language foundation model for remote sensing. *IEEE Transactions on Geoscience and Remote Sensing*, 62:1–16, 2024. 4

[32] Haotian Liu, Chunyuan Li, Qingyang Wu, and Yong Jae Lee. Visual instruction tuning. *Advances in neural information processing systems*, 36:34892–34916, 2023. 2

[33] Haotian Liu, Chunyuan Li, Yuheng Li, and Yong Jae Lee. Improved baselines with visual instruction tuning. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 26296–26306, 2024. 2

[34] Jiang Liu, Hui Ding, Zhaowei Cai, Yuting Zhang, Ravi Kumar Satzoda, Vijay Mahadevan, and R Manmatha. Polyformer: Referring image segmentation as sequential polygon generation. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 18653–18663, 2023. 3

[35] Si Liu, Tianrui Hui, Shaofei Huang, Yunchao Wei, Bo Li, and Guanbin Li. Cross-modal progressive comprehension for referring segmentation. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 44(9):4761–4775, 2021. 7

[36] Sihan Liu, Yiwei Ma, Xiaoqing Zhang, Haowei Wang, Jiayi Ji, Xiaoshuai Sun, and Rongrong Ji. Rotated multi-scale interaction network for referring remote sensing image segmentation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 26658–26668, 2024. 2, 3, 7, 8

[37] Xu Liu and Zhouhui Lian. Rsunivlm: A unified vision language model for remote sensing via granularity-oriented mixture of experts. *arXiv preprint arXiv:2412.05679*, 2024. 3

[38] Ze Liu, Yutong Lin, Yue Cao, Han Hu, Yixuan Wei, Zheng Zhang, Stephen Lin, and Baining Guo. Swin transformer: Hierarchical vision transformer using shifted windows. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 10012–10022, 2021. 5, 6, 8

[39] Siqi Lu, Junlin Guo, James R Zimmer-Dauphinee, Jordan M Nieusma, Xiao Wang, Steven A Wernke, Yuankai Huo, et al. Vision foundation models in remote sensing: A survey. *IEEE Geoscience and Remote Sensing Magazine*, 2025. 2

[40] Xianzhi Ma, Jianhui Li, Changhua Pei, and Hao Liu. Geomag: A vision-language model for pixel-level fine-grained remote sensing image parsing. *arXiv preprint arXiv:2507.05887*, 2025. 3

[41] Ruizhe Ou, Yuan Hu, Fan Zhang, Jiaxin Chen, and Yu Liu. Geopix: A multimodal large language model for pixel-level image understanding in remote sensing. *IEEE Geoscience and Remote Sensing Magazine*, 2025. 2, 3

[42] Chao Pang, Jiang Wu, Jiayu Li, Yi Liu, Jiaxing Sun, Weijia Li, Xingxing Weng, Shuai Wang, Litong Feng, Gui-Song Xia, et al. H2rsvlm: Towards helpful and honest remote sensing large vision language model. *CoRR*, 2024. 3

[43] Chao Pang, Xingxing Weng, Jiang Wu, Jiayu Li, Yi Liu, Jiaxing Sun, Weijia Li, Shuai Wang, Litong Feng, Gui-Song Xia, et al. Vhm: Versatile and honest vision language model for remote sensing image analysis. In *Proceedings of the AAAI Conference on Artificial Intelligence*, pages 6381–6388, 2025. 3

[44] Jerome Quenum, Wen-Han Hsieh, Tsung-Han Wu, Ritwik Gupta, Trevor Darrell, and David M Chan. Lisat: Language-instructed segmentation assistant for satellite imagery. *arXiv preprint arXiv:2505.02829*, 2025. 3

[45] Maryam Rahnemoonfar, Tashnim Chowdhury, Argho Sarkar, Debvrat Varshney, Masoud Yari, and Robin Roberson Murphy. Floodnet: A high resolution aerial imagery dataset for post flood scene understanding. *IEEE Access*, 9:89644–89654, 2021. 2

[46] Hanoona Rasheed, Muhammad Maaz, Sahal Shaji, Abdelrahman Shaker, Salman Khan, Hisham Cholakkal, Rao M. Anwer, Eric Xing, Ming-Hsuan Yang, and Fahad S. Khan. Glamm: Pixel grounding large multimodal model. *The*

*IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024. 6

[47] Nikhila Ravi, Valentin Gabeur, Yuan-Ting Hu, Ronghang Hu, Chaitanya Ryali, Tengyu Ma, Haitham Khedr, Roman Rädle, Chloe Rolland, Laura Gustafson, et al. Sam 2: Segment anything in images and videos. *arXiv preprint arXiv:2408.00714*, 2024. 3, 8

[48] Zhongwei Ren, Zhicheng Huang, Yunchao Wei, Yao Zhao, Dongmei Fu, Jiashi Feng, and Xiaojie Jin. Pixellm: Pixel reasoning with large multimodal model. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 26374–26383, 2024. 3, 6, 7

[49] Esther Rolf, Konstantin Klemmer, Caleb Robinson, and Hannah Kerner. Mission critical–satellite data is a distinct modality in machine learning. *arXiv preprint arXiv:2402.01444*, 2024. 2

[50] Akashah Shabbir, Mohammed Zumri, Mohammed Bennamoun, Fahad S Khan, and Salman Khan. Geopixel: Pixel grounding large multimodal model in remote sensing. *arXiv preprint arXiv:2501.13925*, 2025. 2, 3, 6, 7

[51] Yan Shu, Bin Ren, Zhitong Xiong, Danda Pani Paudel, Luc Van Gool, Begum Demir, Nicu Sebe, and Paolo Rota. Earthmind: Towards multi-granular and multi-sensor earth observation with large multimodal models. *arXiv preprint arXiv:2506.01667*, 2025. 3

[52] Sagar Soni, Akshay Dudhane, Hiyam Debary, Mustansar Fiaz, Muhammad Akhtar Munir, Muhammad Sohail Danish, Paolo Fraccaro, Campbell D Watson, Levente J Klein, Fahad Shahbaz Khan, et al. Earthdial: Turning multi-sensory earth observations to interactive dialogues. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pages 14303–14313, 2025. 3

[53] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. *Advances in neural information processing systems*, 30, 2017. 8

[54] Di Wang, Jing Zhang, Bo Du, Minqiang Xu, Lin Liu, Dacheng Tao, and Liangpei Zhang. Samrs: Scaling-up remote sensing segmentation dataset with segment anything model. In *Advances in Neural Information Processing Systems*, pages 8815–8827, 2023. 3

[55] Fengxiang Wang, Mingshuo Chen, Yueying Li, Di Wang, Haotian Wang, Zonghao Guo, Zefan Wang, Boqi Shan, Long Lan, Yulin Wang, et al. Geollava-8k: Scaling remote-sensing multimodal large language models to 8k resolution. *arXiv preprint arXiv:2505.21375*, 2025. 3

[56] Hao Wang, Limeng Qiao, Zequn Jie, Zhijian Huang, Chengjian Feng, Qingfang Zheng, Lin Ma, Xiangyuan Lan, and Xiaodan Liang. X-sam: From segment anything to any segmentation. *arXiv preprint arXiv:2508.04655*, 2025. 3, 8

[57] Junjue Wang, Zhuo Zheng, Zihang Chen, Ailong Ma, and Yanfei Zhong. Earthvqa: Towards queryable earth via relational reasoning-based remote sensing visual question answering. In *Proceedings of the AAAI conference on artificial intelligence*, pages 5481–5489, 2024. 3

[58] Junjue Wang, Weihao Xuan, Heli Qi, Zhihao Liu, Kunyi Liu, Yuhan Wu, Hongruixuan Chen, Jian Song, Junshi Xia, Zhuo Zheng, et al. Disasterm3: A remote sensing vision-language dataset for disaster damage assessment and response. *arXiv preprint arXiv:2505.21089*, 2025. 2

[59] Zhaoqing Wang, Yu Lu, Qiang Li, Xunqiang Tao, Yandong Guo, Mingming Gong, and Tongliang Liu. Cris: Clip-driven referring image segmentation. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 11686–11695, 2022. 3

[60] Cong Wei, Haoxian Tan, Yujie Zhong, Yujiu Yang, and Lin Ma. Lasagna: Language-based segmentation assistant for complex queries. *arXiv preprint arXiv:2404.08506*, 2024. 3

[61] Cong Wei, Yujie Zhong, Haoxian Tan, Yong Liu, Zheng Zhao, Jie Hu, and Yujiu Yang. Hyperseg: Towards universal visual segmentation with large language model. *arXiv preprint arXiv:2411.17606*, 2024. 3, 8

[62] Cong Wei, Yujie Zhong, Haoxian Tan, Yingsen Zeng, Yong Liu, Zheng Zhao, and Yujiu Yang. Instructseg: Unifying instructed visual segmentation with multi-modal large language models. *arXiv preprint arXiv:2412.14006*, 2024. 2, 3, 5, 8

[63] Chenyun Wu, Zhe Lin, Scott Cohen, Trung Bui, and Subhransu Maji. Phrasecut: Language-based image segmentation in the wild. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 10216–10225, 2020. 6

[64] Zhuofan Xia, Dongchen Han, Yizeng Han, Xuran Pan, Shiji Song, and Gao Huang. Gsva: Generalized segmentation via multimodal large language models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 3858–3869, 2024. 3

[65] Wentao Xiang, Haoxian Tan, Yujie Zhong, Cong Wei, Dengjie Li, and Yujiu Yang. Advancing visual large language model for multi-granular versatile perception. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 22153–22164, 2025. 3

[66] Cilin Yan, Haochen Wang, Shilin Yan, Xiaolong Jiang, Yao Hu, Guoliang Kang, Weidi Xie, and Efstratios Gavves. Visa: Reasoning video object segmentation via large language models. In *European Conference on Computer Vision*, pages 98–115. Springer, 2024. 3

[67] Zhao Yang, Jiaqi Wang, Yansong Tang, Kai Chen, Hengshuang Zhao, and Philip HS Torr. Lavt: Language-aware vision transformer for referring image segmentation. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 18155–18165, 2022. 3, 7

[68] Zhigang Yang, Huiguang Yao, Linmao Tian, Xuezhi Zhao, Qiang Li, and Qi Wang. A large-scale referring remote sensing image segmentation dataset and benchmark. *arXiv preprint arXiv:2506.03583*, 2025. 3

[69] Kelu Yao, Nuo Xu, Rong Yang, Yingying Xu, Zhuoyan Gao, Titinunt Kitrungrotsakul, Yi Ren, Pu Zhang, Jin Wang, Ning Wei, et al. Falcon: A remote sensing vision-language foundation model. *arXiv preprint arXiv:2503.11070*, 2025. 3

[70] Liang Yao, Fan Liu, Hongbo Lu, Chuanyi Zhang, Rui Min, Shengxiang Xu, Shimin Di, and Pai Peng. Remotereasoner: Towards unifying geospatial reasoning workflow. *arXiv preprint arXiv:2507.19280*, 2025. 3, 7

[71] Haobo Yuan, Xiangtai Li, Tao Zhang, Zilong Huang, Shilin Xu, Shunping Ji, Yunhai Tong, Lu Qi, Jiashi Feng, and Ming-Hsuan Yang. Sa2va: Marrying sam2 with llava for dense grounded understanding of images and videos. *arXiv preprint arXiv:2501.04001*, 2025. 3

[72] Zhenghang Yuan, Lichao Mou, Yuansheng Hua, and Xiao Xiang Zhu. Rrsis: Referring remote sensing image segmentation. *IEEE Transactions on Geoscience and Remote Sensing*, 2024. 2, 3, 7, 8

[73] Yang Zhan, Zhitong Xiong, and Yuan Yuan. Skyeyegpt: Unifying remote sensing vision-language tasks via instruction tuning with large language model. *ISPRS Journal of Photogrammetry and Remote Sensing*, 221:64–77, 2025. 2, 3

[74] Pan Zhang, Xiaoyi Dong, Yuhang Zang, Yuhang Cao, Rui Qian, Lin Chen, Qipeng Guo, Haodong Duan, Bin Wang, Linke Ouyang, et al. Internlm-xcomposer-2.5: A versatile large vision language model supporting long-contextual input and output. *arXiv preprint arXiv:2407.03320*, 2024. 7

[75] Shilong Zhang, Peize Sun, Shoufa Chen, Min Xiao, Wenqi Shao, Wenwei Zhang, Yu Liu, Kai Chen, and Ping Luo. Gpt4roi: Instruction tuning large language model on region-of-interest. In *European conference on computer vision*, pages 52–70. Springer, 2024. 2

[76] Wei Zhang, Miaoxin Cai, Tong Zhang, Yin Zhuang, and Xuerui Mao. Earthgpt: A universal multi-modal large language model for multi-sensor image comprehension in remote sensing domain. *IEEE Transactions on Geoscience and Remote Sensing*, 2024. 3

[77] Zheng Zhang, Yeyao Ma, Enming Zhang, and Xiang Bai. Psalm: Pixelwise segmentation with large multi-modal model. In *European Conference on Computer Vision*, pages 74–91. Springer, 2024. 3, 8

[78] Zilun Zhang, Zian Guan, Tiancheng Zhao, Haozhan Shen, Tianyu Li, Yuxiang Cai, Zhonggen Su, Zhaojun Liu, Jianwei Yin, and Xiang Li. Geo-r1: Improving few-shot geospatial referring expression understanding with reinforcement fine-tuning. *arXiv preprint arXiv:2509.21976*, 2025. 3

[79] Yue Zhou, Mengcheng Lan, Xiang Li, Litong Feng, Yiping Ke, Xue Jiang, Qingyun Li, Xue Yang, and Wayne Zhang. Geoground: A unified large vision-language model for remote sensing visual grounding. *arXiv preprint arXiv:2411.11904*, 2024. 2, 3, 7

[80] Minjie Zhu, Yichen Zhu, Xin Liu, Ning Liu, Zhiyuan Xu, Chaomin Shen, Yaxin Peng, Zhicai Ou, Feifei Feng, and Jian Tang. Mipha: A comprehensive overhaul of multi-modal assistant with small language models. *arXiv preprint arXiv:2403.06199*, 2024. 6

[81] Yongshuo Zong, Qin Zhang, Dongsheng An, Zhihua Li, Xiang Xu, Linghan Xu, Zhuowen Tu, Yifan Xing, and Onkar Dabeer. Ground-v: Teaching vlms to ground complex instructions in pixels. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pages 24635–24645, 2025. 3